25900060 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION, MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) -

MG2DSCECT100,MG2DSCECC100 - ESSENTIAL CONCEPTS IN DIGITAL ELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. What is the binary multiplication of 110 * 11 [U] / [CO1]

   a). 10010 b). 11001

   c). 10101 d). 10110

2. What is the significance of NAND/NOR gates in CMOS design? [K] / [CO2]

   a). Power saving b). Area efficiency

   c). Universal functionality d). Speed

3. Subtract the signed numbers $(00010011)_2$ from $(11100111)_2$ [U] / [CO1]

   a). $(11010111)_2$ b). $(11010100)_2$

   c). $(11010110)_2$ d). $(11010101)_2$

4. Which of the following is an example of a digital device? [K] / [CO1]

   a). Thermometer b). Analog clock

c). Digital watch d). Loudspeaker

5. 1+A= [U] / [CO2]

a). 1 b). 0 c). A d). $\overline{A}$

6. What is the base of the binary number system? [K] / [CO1]

a). 2 b). 10 c). 8 d). 16

7. Add 1100 and 1010. [U] / [CO1]

a). 10011 b). 11010

c). 10110 d). 11100

8. Which Boolean operation is used to combine the sum terms in POS [U] / [CO2]

a). AND b). OR

c). XOR d). NOT

9. What is the primary function of a Half Adder? [K] / [CO3]

a). Adds two binary digits b). Subtracts two binary digits

c). Multiplies two binary digits d). Divides two binary digits

10. What is the main difference between an SR latch and an SR flip-flop? [K] / [CO3]

a). The SR latch is edge-triggered, while the SR flip-flop is level-triggered

b). The SR latch is level-triggered, while the SR flip-flop is edge-triggered

c). The SR latch has two inputs, and the SR flip-flop has three inputs including clock

d). There is no difference; they are the same

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Convert the following binary numbers into decimal form: (a) 111011 (b) 110111. [U] / [CO1]

12. Use a K-map to find the minimum POS form for the expression [U] / [CO2]

$$(A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(A+\bar{B}+\bar{C})+(\bar{A}+\bar{B}+C)$$

13. What are the applications of the JK flip-flop in digital circuits? [K] / [CO3]

14. Describe the working of a NOR gate with truth table. [K] / [CO2]

15. Write the truth table for a 1:4 demultiplexer and explain its function. [K] / [CO3]

16. Convert the following hexadecimal numbers to binary: (a)(57)$_{16}$ (b) (3A5)$_{16}$ (c) (F80B)$_{16}$. [K] / [CO1]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Define the following logic gates i)AND gate ii) OR gate iii) NOT gate iv) XOR gate For each gate, provide the truth table, logic symbol, and Boolean expression. [U] / [CO2]

18. Explain SISO shift register in detail [K] / [CO3]

19. With the help of suitable examples explain two forms of DeMorgan's Theorem [U] / [CO2]

20. i) Convert the hexadecimal numbers 7A5B and 82BC to Decimal. (2 Marks) [K] / [CO1]
    ii) Convert the octal numbers 3143 and 3475 to decimal. (2 Marks)
    iii) Convert the decimal numbers 4096 and 5678 to hexadecimal. (2 Marks)
    iv) Convert the binary numbers 111000 to octal. (2 Marks)
    v) Convert the hexadecimal numbers 4CDA and 89FB to decimal. (2 Marks)

[10x2 = 20]